

Youyin Hu

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EDUCATION

2024.09 – Present	University of Shanghai for Science and Technology Operations Research and Control Theory	Master of Science GPA: 4.0/4.5
Relevant Courses: Numerical Methods for Differential Equations (Grade: 98/100), Statistical Learning Theory (Grade: 97/100), Graph Neural Networks (Grade: 93/100), Stochastic Control (Grade: 91/100)		
2020.09 – 2024.06	University of Shanghai for Science and Technology Applied Mathematics	Bachelor of Science
Relevant Courses: Ordinary Differential Equations (Grade: 93/100), Probability Theory (Grade: 95/100), Abstract Algebra (Grade: 97/100), Machine Learning (Grade: 90/100), Fuzzy Mathematics (Grade: 95/100)		

RESEARCH EXPERIENCE

2024.06 - Present	Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme	Independent Researcher
- Conceptualized and developed a novel homomorphic encryption-based push-sum framework to bridge the gap between data privacy and communication efficiency in multi-agent systems. - Designed a unified protocol integrating Paillier encryption and encoding-decoding quantization, effectively handling asymmetric directed communication topologies. - Derived rigorous theoretical convergence guarantees for uniformly strongly connected time-varying directed graphs, establishing explicit convergence rate bounds. - Formulated a comprehensive error propagation analysis, proving that quantization errors decay exponentially to ensure optimal solution convergence. - Authored the manuscript: "Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme" (<i>In preparation</i>).		
2024.09 – 2025.02	Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects	Independent Researcher
- Developed a distributed recursive filter for state-saturated systems, utilizing output mask functions to ensure data privacy and scaled uniform quantization to optimize communication efficiency in sensor networks. - Derived the upper bound of filtering error covariance via Riccati-like equations, proving its boundedness over a finite horizon and recursively solving for the optimal gain matrix to balance accuracy and stability. - Verified the algorithm's robustness against nonlinear saturation through a three-tank system simulation in MATLAB. - Published findings in International Journal of Network Dynamics and Intelligence (2025, DOI: 10.53941/ijndi.2025.100012).		
2025.09	The 22nd China Graduate Mathematical Contest in Modeling	Team Leader
- Led modeling efforts in a 3-person team, developing turbulence intensity calculation models using Richardson number and TKE methods with multi-source data fusion. - Implemented 3D spatial turbulence mapping algorithms and prediction models with 100m horizontal and 50m vertical resolution; performed optimal flight route planning using Python and MATLAB with comprehensive visualizations. - Completed research paper and computational framework within 72 hours, awarded National Third Prize for innovative data fusion approach and practical solutions for low-altitude flight safety.		

2025.02 General Program of National Natural Science Foundation of China Modeling Lead

- Led mathematical modeling for distributed optimization problems with communication constraints, developing novel neural network architectures integrating binary and multi-description coding schemes for resource-limited multi-agent systems.
- Collaborated with team members on homomorphic encryption strategies for privacy- preserving distributed learning, contributing to theoretical framework development and experimental validation through numerical studies.
- Designed adaptive control algorithms addressing packet loss, bandwidth constraints, and model uncertainty, validated performance through MATLAB simulations and numerical analysis.

2023.12 – 2024.05 Research on Linearizable Initial Boundary Value Problems for Sasa-Satsuma Type Equations Lead Researcher

- Investigated linearizable boundary value problems for generalized Sasa-Satsuma equations on the half-line using Riemann-Hilbert problem approach; constructed 3×3 matrix Riemann- Hilbert formulation with spectral functions defined through Lax pairs.
- Derived jump matrices and residue conditions for specific linearizable boundary conditions, demonstrating that spectral functions can be computed through algebraic manipulation of global relations without solving nonlinear equations; utilized **Maple** for symbolic computation and verification of complex algebraic expressions.
- **Awarded Outstanding Dissertation** for developing a comprehensive theoretical framework that simplifies the reconstruction process of initial-boundary value problems, contributing to the unified transform method for integrable partial differential equations.

RESEARCH INTEREST

- Distributed optimization and multi-agent systems
- Privacy-preserving computation and homomorphic encryption
- Networked control systems

PUBLICATIONS

- Hu, Y.; Zhang, C.; Liu, S. Privacy-Preserving Distributed Recursive Filtering for State-Saturated Systems with Quantization Effects. International Journal of Network Dynamics and Intelligence 2025, 4 (2), 100012. <https://doi.org/10.53941/ijndi.2025.100012>
- Hu, Y.; Liu, S.; Wang, L. Push-Sum Distributed Optimization for Multi-Agent Systems Under Encoding-Decoding Scheme. (In preparation)

TEACHING EXPERIENCE

Teaching Assistant, Stochastic Control University of Shanghai for Science and Technology Spring 2025

- Assisted Prof. Shuai Liu with course instruction for graduate-level stochastic control.
- Delivered tutorial sessions on control theory fundamentals and applications.

SKILLS

Programming & Analysis: Python (NumPy, SciPy, SymPy, Matplotlib), SQL

Mathematical Software: MATLAB, Mathematica, Maple

Scientific Writing: LaTeX

Language Proficiency: Chinese (Native), English (IELTS 7.0)